MISSION: POWER UP THE FHR

An Introduction to Reactor Dynamics

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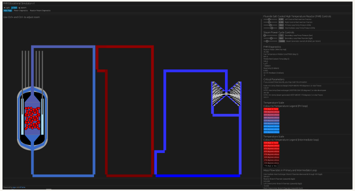
May 18, 2026

MISSION BRIEFING: GRID DEMAND INCREASE

Currently, our Fluoride-salt-cooled High-temperature Reactor (FHR) is operating stably, supplying baseload power to the grid.

The Situation: Grid demand has just spiked. The grid operator requests an immediate increase in power output.



Your Objective

Increase reactor thermal power from **30 MWth** to **60 MWth** and stabilize it there.

OPERATIONAL CONSTRAINTS & SAFETY LIMITS

Nuclear reactors are powerful systems that require precise control. We cannot just ”floor the accelerator.”

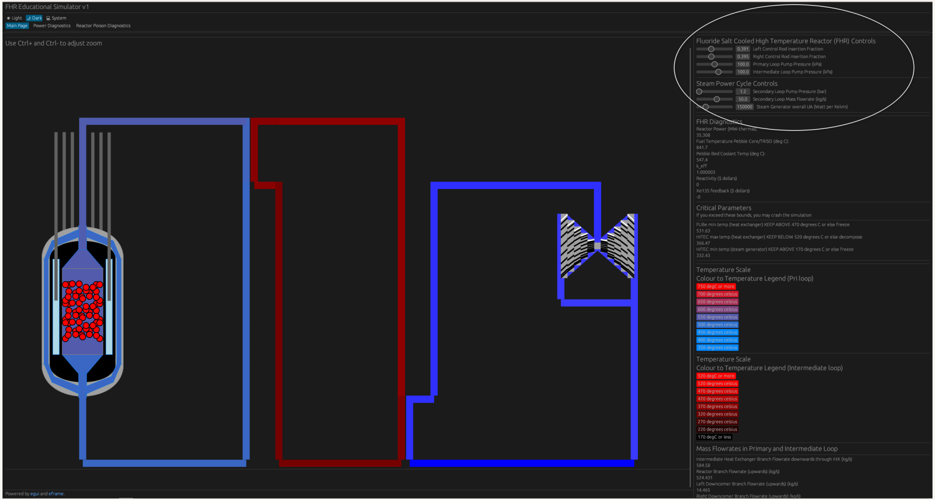
The Rules of Engagement:

- You have full control over the reactor’s control rods (reactivity).
- You must aim for exactly **60 MWth**.

SAFETY LIMIT (FAILURE CONDITION)

If reactor power exceeds **1000 MWth** at any point during the maneuver, you have violated the reactor’s safety limits and failed the mission.

YOUR INTERFACE: THE FHR CONTROL PANEL



Current State:

Stable at 30 MWth

Volunteers Needed!

Who thinks they have the steady hand to bring us to 60 MWth without exceeding 1000 MWth?

For the rest of the class (Observers): Watch closely. Don't just look at the power level.

- How does the reactor respond immediately after they move the controls?
- What happens to the power when they *stop* moving the controls?
- Does the reactor feel "heavy" (slow to move) or "twitchy" (fast to move)?

DEBRIEF: INTUITION VS. REALITY

We just saw a few attempts. It wasn't as easy as turning up a volume knob on a radio.

Discussion Questions:

1. Did anyone overshoot the 60 MWth target? Why did that happen?
2. Did you notice the power continuing to change even after you put the controls back to neutral?
3. If you added too much reactivity, did it feel like the reactor was "running away" from you?

Why does the reactor have "momentum"? Why doesn't it stop instantly?
To answer this, we need to look at the physics of neutron populations.

THE HIDDEN MECHANISM

You experienced the reactor's behavior:

- It doesn't respond instantly to control rod changes.
- It has "momentum" – power keeps changing after you stop acting.
- If you add too much reactivity quickly, it feels uncontrollable.

To understand why, we need to look at the life cycle of neutrons.
We use the **Point Reactor Kinetics Equations (PRKE)** to model this.

STEP 1: THE NEUTRON BALANCE

At its simplest, reactor power changes based on the population of neutrons (n) Bell and Glasstone 1970.

$$\text{Rate of Change} = \text{Production} - \text{Loss}$$

$$\frac{dn}{dt} = \text{Production} - \text{Loss}$$

The production and loss terms depend on many nuclear processes:

- Fission reactions (production)
- Absorption in fuel, coolant, structure (loss)
- Leakage out of the core (loss)
- Fast vs thermal neutron behavior

How do we capture all these effects in one framework?

THE SIX FACTOR FORMULA

Nuclear engineers use the **six factor formula** to account for all the ways neutrons are gained and lost in a reactor Lamarsh, Baratta, et al. 2001:

$$k_{eff} = \underbrace{P_{TNL} \cdot P_{FNL}}_{\text{Non-Leakage}} \cdot \underbrace{\eta}_{\text{Reproduction}} \cdot \underbrace{\epsilon}_{\text{Fast Fission}} \cdot \underbrace{f}_{\text{Thermal Utilization}} \cdot \underbrace{p}_{\text{Resonance Escape}}$$

Where:

- P_{TNL}, P_{FNL} = Probability of not leaking (thermal and fast)
- η = Neutrons produced per neutron absorbed in fuel
- ϵ = Fast fission factor
- f = Thermal utilization factor
- p = Resonance escape probability

WHAT DOES k_{eff} TELL US?

The effective multiplication factor k_{eff} represents the ratio of neutrons in one generation to the next:

$$k_{eff} = \frac{\text{Neutrons in Generation } (n+1)}{\text{Neutrons in Generation } n} = \frac{\text{Production}}{\text{Loss}}$$

This leads to three reactor states:

- $k_{eff} < 1$: **Subcritical** – neutron population decreasing
- $k_{eff} = 1$: **Critical** – neutron population constant
- $k_{eff} > 1$: **Supercritical** – neutron population increasing

Now we can rewrite our balance equation:

$$\frac{dn}{dt} = \text{Loss} \cdot (k_{eff} - 1)$$

STEP 2: WHAT IF ALL NEUTRONS WERE FAST?

Imagine every neutron is born instantly after fission (”prompt neutrons”).
The ”Loss” term depends on how quickly neutrons are removed. We define the **neutron lifetime** (l) as the average time a neutron survives before being lost (absorbed or leaked) Lamarsh, Baratta, et al. 2001.

$$\text{Loss Rate} = \frac{n}{l}$$

Substituting this into our balance equation gives the **Prompt Kinetics Equation**:

$$\frac{dn}{dt} = \frac{n}{l}(k_{eff} - 1)$$

This equation assumes *instantaneous* response.

HOW FAST ARE PROMPT NEUTRONS BORN?

Prompt neutrons are emitted virtually instantaneously after a fission event occurs.

Timescale for prompt neutron emission Mihalczo 2004:

$$t_{\text{prompt emission}} \approx 10^{-14} \text{ seconds}$$

This is **10 femtoseconds** – essentially instantaneous on any human or engineering timescale.

Once born, these neutrons then live for a time period l (the neutron lifetime) before being absorbed or leaking out. This lifetime l determines how quickly the reactor responds.

How fast is this response?
The neutron lifetime l depends on the reactor type Lamarsh, Baratta, et al. 2001:

Reactor Type	Neutron Energy	Neutron Lifetime (l)
Thermal (LWR, FHR)	$\sim$0.025 eV	10^{-4} to 10^{-5} s
Fast (SFR, metal core)	$\sim$1 MeV	10^{-7} to 10^{-8} s

Fast reactors respond 1000× faster than thermal reactors!

If $k_{eff} = 1.001$ in a fast reactor, power doubles in microseconds.

Reality Check

Human reaction time $\approx$ 0.25 seconds = 250,000 microseconds.
Prompt-only reactors would be **impossible** for humans to control.

STEP 3: NATURE'S SAFETY BRAKE - DELAYED NEUTRONS

Thankfully, not all neutrons are born instantly Bell and Glasstone 1970.
A small fraction, denoted by β (beta), are emitted by fission fragments (called **precursors**, C) seconds or even minutes later.

- **Prompt Neutrons ($1-\beta$):** Appear instantly ($\sim 10^{-14}$ s).
- **Delayed Neutrons (β):** Appear later, governed by the decay of precursors (seconds to minutes).

This delay is what creates the "momentum" or lag you felt in the simulator. The precursors act like a battery, storing potential neutrons and releasing them slowly.

THE POINT REACTOR KINETICS EQUATIONS (1-GROUP)

To model reality, we need two coupled equations: one for neutrons (n) and one for precursors (C) Bell and Glasstone 1970.
We introduce **Reactivity (ρ)** and **Generation Time (Λ)** to simplify the math.

The 1-Group PRKE:

$$\underbrace{\frac{dn}{dt}}_{\text{Change in Power}} = \underbrace{\frac{\rho-\beta}{\Lambda}n}_{\text{Prompt Term}} + \underbrace{\lambda C}_{\text{Delayed Source}}$$

$$\underbrace{\frac{dC}{dt}}_{\text{Change in Precursors}} = \underbrace{\frac{\beta}{\Lambda}n}_{\text{New Precursors}} - \underbrace{\lambda C}_{\text{Precursor Decay}}$$

Where λ is the precursor decay constant, and reactivity is defined as:

$$\rho \equiv \frac{k_{eff}-1}{k_{eff}}$$

THE EDGE OF CONTROL: PROMPT CRITICALITY

Look closely at the prompt term in the neutron equation:

$$\frac{\rho - \beta}{\Lambda} n$$

Normal operation involves adding small amounts of reactivity ($\rho < \beta$). The prompt term is negative, and the reactor relies on the delayed source (λC) to increase power slowly.

What happens if you add so much reactivity that $\rho > \beta$?

Prompt Critical

The prompt term becomes **positive**. The reactor power begins to rise exponentially based *only* on prompt neutrons, on a timescale of microseconds.
Control is lost instantly.

UNITS OF REACTIVITY: THE DOLLAR

Because the fraction β is the boundary between controllable and uncontrollable, we use it as a unit.

$$\text{Reactivity in Dollars (\$)} = \frac{\rho}{\beta}$$

- **\$0.10 (10 cents):** A typical, safe control movement.
- **\$0.50 (50 cents):** A large, but manageable insertion.
- **\$1.00 (Prompt Critical):** The exact boundary where $\rho = \beta$.
- **> \$1.00 (Super-Prompt Critical):** The danger zone.

For typical reactors: $\beta \approx 0.0065$ (U-235) to 0.0021 (Pu-239).
So \$1.00 $\approx$ 650 pcm (percent-mille) to 210 pcm of reactivity.

CHALLENGE REDUX: NOW MONITOR REACTIVITY IN DOLLARS

Now that you understand reactivity in dollars, let's try the power maneuver again.

Updated Mission: 30 MWth → 60 MWth

This time, the FHR simulator displays **reactivity in dollars ($)** on the interface.

New constraint: Never exceed **$0.50** during the maneuver.

Observers: Watch for:

- How much reactivity does the operator insert?
- How does the reactor respond at different reactivity levels?
- What happens when reactivity returns to zero?

Who wants to try with this new information?

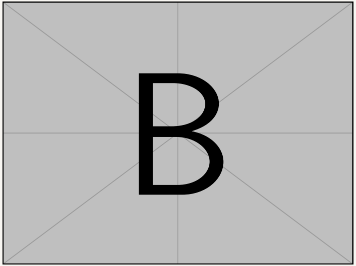
HISTORICAL WARNING: THE SLOTIN INCIDENT (1946)

Los Alamos National Laboratory, May 21, 1946
Stratton and Smith 1989.

Physicist Louis Slotin was demonstrating a criticality experiment to his replacement. The experiment involved bringing two hemispheres of plutonium (the ”Demon Core”) close together.

The only thing keeping them apart? A screwdriver blade.

The procedure was unauthorized and extremely dangerous. Colleagues called it ”tickling the dragon’s tail.”



Recreation of the ”Demon Core” experiment setup.

THE SLOTIN INCIDENT: THE ACCIDENT

At 3:20 PM, Slotin's screwdriver slipped Stratton and Smith 1989; Oettingen 2018.

The hemispheres came together. The core went **super-prompt critical** instantly—estimated at roughly **$1.10** Oettingen 2018.

What happened in that instant:

- Bright flash of blue light (Cherenkov radiation)
- Wave of heat across Slotin's hand
- Sour taste in his mouth (ionization of air)
- Duration: less than one second

Slotin immediately knocked the top hemisphere away with his bare hand, stopping the reaction. His quick action likely saved the lives of the seven other people in the room.

But the damage was done.

THE SLOTIN INCIDENT: THE AFTERMATH

Radiation dose received: Estimated 2,100 rad (21 Gy) in less than one second Stratton and Smith 1989.

Timeline of Louis Slotin's condition:

- **Day 1:** Nausea and vomiting began within hours
- **Day 2-3:** Severe radiation burns appeared on his hands
- **Day 4-6:** Mental confusion, loss of coordination
- **Day 7-8:** Internal organs failing, extreme pain
- **Day 9:** Louis Slotin died on May 30, 1946

The Critical Lesson

Human reaction time ($\sim$0.25 s) is meaningless when prompt criticality unfolds in microseconds. Even Slotin's heroic reflexes couldn't save him—he had already received a lethal dose before his brain could process what happened.

THE BORAX EXPERIMENTS: PROVING THE DANGER (1950S)

After incidents like Slotin's, the nuclear community needed to understand prompt criticality in *reactor* systems, not just bare cores.

The Question: What happens if a water-cooled reactor goes prompt critical?

The Boiling Water Reactor Experiments (BORAX) at Idaho National Laboratory were designed to answer this—by deliberately destroying reactors.

The BORAX-I Final Test (1954)

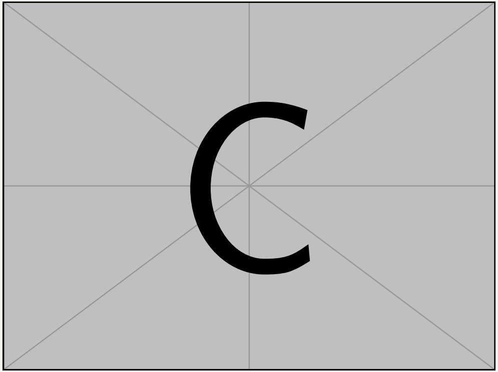
Scientists programmed the reactor to eject a control rod rapidly, inserting reactivity well above $1.00.
They evacuated everyone to a safe distance and filmed what happened.

BORAX-I: THE DESTRUCTION

The sequence of events:

1. Control rod ejected
2. Reactor power rose exponentially (milliseconds)
3. Fuel temperature skyrocketed
4. Water flash-boiled to steam
5. Steam explosion launched the reactor vessel into the air
6. Complete destruction of the reactor

[Watch the BORAX destruction test]



BORAX-I final destructive test—reactor vessel airborne.

BORAX: WHY DIDN'T SAFETY FEATURES WORK?

Boiling Water Reactors (BWRs) have multiple passive safety features:

- Natural circulation cooling
- Negative void coefficient (steam bubbles reduce reactivity)
- Negative fuel temperature coefficient

So why did the reactor explode?

The Speed Problem

Prompt criticality happens in **milliseconds**. The fuel temperature feedback *did* activate and stop the nuclear reaction—but not before the fuel temperature rose so high that it **melted the fuel**.
The molten fuel then violently interacted with water, causing a steam explosion.
Physics worked—but too slowly to prevent damage.

CAN MODERN REACTORS SURVIVE PROMPT CRITICALITY?

It depends on the reactor design and fuel type.

Light Water Reactors (LWRs) / BWRs

Negative fuel temperature coefficient stops the reaction quickly.
However, the fuel temperature can rise so high (>2800°C) that the fuel **melts** before the feedback stops the excursion.
Result: Fuel damage, potential steam explosion (as in BORAX).

Key issue: Low thermal margin in UO_2 fuel—melting point is relatively close to operating temperature.

HIGH-TEMPERATURE REACTORS: BETTER MARGINS

HTGRs (High-Temperature Gas Reactors) and FHRs

These reactors use **TRISO fuel particles**—ceramic-coated fuel designed to withstand extreme temperatures (>1600°C).

Key advantages:

- **Higher thermal margins:** TRISO fuel can tolerate much higher temperatures before failure
- **Strong negative temperature coefficient:** Fuel temperature rise quickly reduces reactivity
- **Lower excess reactivity:** Less potential for large reactivity insertions

Result: Fuel temperature feedback *may* stop a prompt criticality event without damaging the fuel—**provided the excess reactivity insertion isn't too large.**

This is why HTGRs and FHRs are considered to have superior passive safety

THE BOTTOM LINE: NEVER GO PROMPT CRITICAL

Regardless of reactor type, prompt criticality is unacceptable.

- **LWRs/BWRs:** Prompt criticality leads to fuel melting and potential steam explosions
- **HTGRs/FHRs:** May survive small prompt criticality events, but only if excess reactivity is limited
- **Fast reactors:** Even more sensitive due to shorter neutron lifetimes

The Engineering Rule

All reactors are designed with **reactivity limits** to ensure they can *never* reach \$1.00 under any credible scenario.
Multiple independent systems (control rods, shutdown systems, reactivity coefficients) provide defense-in-depth.

This is why you must understand and respect the dollar boundary.